Hui-Jou Chou

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years of experience designing and delivering full-stack web applications, cloud infrastructure, and data pipeline solutions. Specialized in Python, React, and AWS with a track record of reducing manual effort and accelerating workflows through automation. Proven ability to independently architect end-to-end systems — from serverless cloud deployments to analytics dashboards — and deliver measurable business impact. Seeking a Senior Software Engineer role in data-driven or cloud-native product teams.

EXPERIENCE

Research Systems Specialist (Software Engineer) | ETS — *Jan 2021 – Present*

NIA-NAEP Team | Senior Track

- **Cloud Architecture:** Independently designed and deployed a private serverless Data Viewer App using AWS S3 as frontend (Next.js), Lambda, API Gateway (Python), and RDS for SQL Server within a secured VPC — eliminating 100% of infrastructure overhead and automating 95% of build/test/deploy workflows via CodePipeline and CodeBuild.
- **Full-Stack Engineering:** Engineered a SQL Server administration tool using React and Python, enabling automated backups, restores, data integrity checks, and high-volume bulk data loads — streamlining DBA workflows and reducing manual intervention significantly.
- **Data & Analytics:** Engineered a custom ETL pipeline and Power BI analytics dashboard integrated with Copilot Studio data agents, streamlining data workflows and delivering automated, actionable business insights.

Associate Research Systems Specialist | ETS — *Jan 2018 – Jan 2021*

NIA-NAEP Team

- **Document Automation:** Built a Flask + jQuery Document Assembly Tool (DAT) that auto-generated data-driven PDF reports with dynamic tables, images, and graphics — reducing manual document preparation time by 70–80%. Added a CMS interface and Excel export to further streamline content management.
- **Accessibility & Reusability:** Co-developed the DDTV (Data-Driven Table Visualization) JavaScript library, designing the core table-generation engine that produces fully Section 508–compliant tables — reducing manual construction effort by 50–70% and enabling adoption across 5+ NAEP reporting projects.
- **Data Visualization:** Developed interactive SVG charts for large-scale educational assessment reports (PIAAC and NAEP) using Billboard.js and HighchartsJS, delivering reusable visualization solutions across 50+ states and districts to make complex assessment data accessible to stakeholders.

Software Developer Intern | Microdysis Inc. — *Dec 2016 – Jul 2017*

Medical Device Team

- Developed a Windows C application using Win32 and Silicon Labs Bluetooth API to read/write data to a smoking detection device over Bluetooth.
- Built an Android Java app connecting to a heart rate monitor via Bluetooth and uploading data to Dropbox; contributed new features to a stress management web app using Bootstrap and AngularJS.

Research Assistant | Dept. of Biology, Rutgers University — *Sep 2015 – May 2017*

Bioinformatics Lab

- **Algorithm Development:** Developed a parallel-computing algorithm (C, pthreads, SAMtools) to detect DNA structural variants from Pacific Biosciences long-read genome sequencing data — achieving higher sensitivity and precision than other published SV detection tools when benchmarked against human genome data.
- **Statistical Analysis:** Performed algorithm performance analysis using Python Pandas, producing benchmarking reports that validated the tool's superiority over existing solutions.

EDUCATION

M.S., Computer Science — Rutgers University–Camden *May 2017 | GPA 3.6/4.0*
M.S., Biochemistry — National Taiwan University, Taiwan *Jun 2011 | GPA 3.6/4.0*

TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript, Java, C, Swift, SQL
Frontend: React, Angular, HTML, CSS, jQuery, Highcharts, D3
Backend: Flask, Django
Cloud & DevOps: AWS (Lambda, ECS, EC2, RDS, Amplify, API Gateway, S3, CodePipeline, CodeBuild, Step Functions, SNS), Docker
Data & Analytics: SQL Server, Pandas, Power BI
AI / ML: LangChain, Copilot Studio, Github Copilot, OpenCode, Claude Code